

CHE 205, Fall 2019, Problem Set 3

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Due date: 6:00pm, Fri, 09/20/2019

All problems are from the textbook by V.L. Law.

Submit 2 files on blackboard. For each problem, submit 1 file. Create different tabs (sheets) for different parts (a, b, c, etc.) of the same problem.

Problem 1 (10 pts). Exercise 2.4.

Problem 2 (10 pts). Exercise 2.6.

Exercise 2.4: Design and Code a VBA program to perform the following tasks:

- a. Generate a list of 20 numbers between 1 and 100 using the Excel function `RANDBETWEEN`. Select these numbers and copy them (by value) to another column (the reason for this is that `RANDBETWEEN` updates the list each time an operation is performed in the spreadsheet).
- b. Input the list of numbers generated in part a. Any method can be used to indicate the “end of data,” such as a blank cell as used in Example 1.1.
- c. Compute the following “statistics” for the input numbers:
 1. Average
 2. Standard deviation
 3. Largest in absolute value
 4. Smallest in absolute value

Necessary formulas are as follows:

$$\text{average} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{standard deviation} = s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Check your results by using the Excel functions `AVERAGE`, `STDEV`, `MAX`, and `MIN`.

Exercise 2.6: Sorting is a frequently used operation in many data processing applications. Excel has a powerful sorting tool, available from the Data/Sort menu. If it is desired to sort data under control of a VBA program, such code might need to be written. The so-called bubble sort algorithm is not a very efficient one, but it is easy to program and it easily will suffice for small data sets. The essence of the bubble sort method is as follows: Repeatedly step through the list to be sorted, comparing two items at a time and swapping them if they are in the wrong order. The process is repeated until no swaps are needed, which indicates that the list is sorted. The algorithm gets its name from the way smaller elements (when sorting in a descending order) “bubble” to the top of the list. The following is a *pseudo-code* description of the process. Pseudo-code looks similar to VBA code, but no particular attention is paid to strict coding rules. The pseudo-code assumes that the array name is A and its length is N.

```
swapped = true           'swapped is a logical (boolean) variable
while swapped
  swapped = false        'temporarily assume the list is sorted
  for i = 1 to N - 1
    if A(i) > A(i+1) then
      swap A(i) and A(i+1)
      swapped = true      'a swap indicates an unsorted list
    end if
  end for
end while
```

This logic could just as easily have been illustrated with a structure chart.

Generate a list of 20 numbers between 1 and 100 using the Excel function `RANDBETWEEN`. Select these numbers and copy them (by value) to another column (the reason for this is that `RANDBETWEEN` updates the list each time an operation is performed in the spreadsheet). Write a VBA program to input the copied numbers and sort them into descending order using bubble sort. Output the sorted numbers to the spreadsheet. Be sure to give an error message if no numbers are entered.